Companion in the Boundary Problem network. Build companions: How to Build a UFO · The DIY Hoverbike. History companion: The Military-Industrial Complex · UAP Origins.

Antigravity Devices

What actually lifts, what is only claimed, and what the physics forbids — a builder's field guide to the two real levitation effects, the redacted scientists, and the "it also makes energy" claims.

Osika Innovation AB · Alexander Osika, Claude — draft 0.3, 2026-08-18 (draft 0.1, 2026-06-14; draft 0.2 same day: §8 and the Schauberger table row corrected on the evidence of `viktor_schauberger.html` draft 0.6 — "pseudoscience" withdrawn; draft 0.3: Otis Carr type-specimen box in §9, table row and references, on the evidence of `otis_carr.html` v1.0; §9 and the §10 over-unity row re-registered from "forbidden" to the energy lane's first-principles reservoir test — Moddel-class open-system extraction now stated as real and measured, per the 2026-06-20 open-system doctrine this draft had predated).

> *"I do not think there is any thrill that can go through the human heart like that felt by the inventor as he sees some creation of the brain unfolding to success."*
>
> — Nikola Tesla

> *"Put strong limits on all proposed theories."*
>
> — M. Tajmar & M. Schreiber, title of the 2021 vacuum-thrust null that settles the electrogravitics question

*Two levitation effects are real, cost under a hundred dollars, and lift small masses this weekend on textbook physics: **acoustic radiation pressure** (an ultrasonic standing wave that floats beads, droplets, and insects) and **electrohydrodynamic ion thrust** (a high-voltage "lifter" that pushes against air). Almost every "antigravity device" in the historical record is one of these two effects — usually the second — or a theory that was refuted, or a claim that was never independently replicated. Both real effects have hard physical ceilings that forbid a human-carrying free-air version. The "it also generates energy" layer — Tesla's earth power, Schauberger's implosion, the self-powered disc — rests on genuine but modest physics (Schumann resonance, the global atmospheric circuit, the Casimir force, vortex aeration and micro-hydro) inflated into conservation-law violations that have never survived a closed-loop measurement — and, in Schauberger's case, were never measured at all. This paper gathers all of it, sorts it by what the boxes below mean, and ends with the only honest move: build the two demonstrators, run the controls, and let the receipt — not the legend — speak.*

How to read the boxes. This is a build guide and an honest survey at the same time. The colored boxes are the instrument that keeps those two jobs from contaminating each other.

- **Documented** — on the public or scientific record; a patent number, a journal DOI, an obituary; checkable without trusting us.
- **Build it now** — you can construct this and it will work on conventional physics; real performance numbers, not aspirational ones.
- **Proponent's claim** — asserted by an inventor or advocate and not independently verified. Not a synonym for "false" — a synonym for "unconfirmed."
- **Physics verdict** — what a controlled experiment or settled theory actually concluded, including the honest nulls. This is where signal gets separated from noise.
- **Open thread** — a lead that is genuinely unresolved and worth pulling. Named so you can pull it too.
- **Safety** — a real hazard. Lethal voltages and lung-rupturing sound are both in this paper.

One discipline holds the document together, inherited from the rest of this network: **name what is buildable, name what is claimed, and let the measurement — not the wish — promote a claim across the line.**

Where this sits. `how_to_build_a_ufo.html` gives the framework definition of an anti-gravity-class effect (a controlled *settling bias* of the consensus lattice, not a conservation-law violation) and reverse-engineers the saucer. `diy_hoverbike_paper.html` is the four-cymbal vehicle. This paper is the bench layer beneath both: the devices small enough to hold in your hand, the people who chased the effect and what became of them, and a clean ledger of which claims the physics licenses. It corrects two over-reaches in the companions where the honest ceiling demands it (§2.4, §3.5).

§1 The honesty contract

"Antigravity" is three different claims wearing one word, and the fastest way to waste a year is to let them blur. This paper keeps them apart from the first page.

1. **Lift on conventional physics, now.** A measurable upward force on an object, produced by a mechanism that is fully accounted for by acoustics or electrohydrodynamics. No new physics. This is real, and most of Part I is how to build it.
2. **A lattice-coupling / x_v effect, under test.** The framework claim of the companion papers: a coherent body biases how the consensus lattice settles, producing a force that conventional bookkeeping of the same hardware does *not* predict (`how_to_build_a_ufo.html` §4). This is not yet a receipt. It is a hypothesis with a pre-registered bench test, and it is allowed to stay open — but it is not allowed to borrow credibility from category 1.
3. **Free-space, self-powered, conservation-defying antigravity.** The sci-fi version, and the version most of the historical claims quietly assume. This is the one the physics forbids, and Part III is where it goes to be weighed.

> Thesis sentence · *The contract: a device may be celebrated as buildable only when the mechanism is conventional and the numbers are real; a device may be called an antigravity candidate only after it has beaten the controls that separate it from category 1; and no device gets to be both "lifts" and "makes free energy" without showing a closed-loop measurement, because that pair is exactly what conservation of energy forbids.*

§2 The acoustic track — ultrasound and cymbals

The cleanest hand-held levitation you can build. A standing sound wave has pressure nodes; a small dense object dropped into one feels a time-averaged *acoustic radiation pressure* that pushes it toward the node and holds it against gravity. The trapping potential is Gor'kov's; the force scales with the acoustic energy density and with the object's volume, which is the whole story of what it can and cannot lift.

```
top array   ▼▼▼▼▼▼▼▼   36 × 40 kHz transducers
              |   N   |        λ = c/f = 343/40000 = 8.6 mm
              |  (·o·) |  ← bead trapped at a pressure NODE
              |   A   |        node spacing = λ/2 ≈ 4.3 mm
              |   N   |        object must be smaller than λ/2 (< ~4 mm)
bottom array ▲▲▲▲▲▲▲▲   36 × 40 kHz, driven in ANTIPHASE
```

Figure 1. Single-axis standing-wave levitator (TinyLev topology). Two opposed phased arrays focus to a common point; objects park at the nodes.

2.1 The buildable device

> **Build it now** · **TinyLev** (Marzo, Barnes & Drinkwater, *Rev. Sci. Instrum.* 88, 085105, 2017) is the open-source standard and the one to build first.
>
> - **Emitters:** 72 ultrasonic transducers (10 mm, 40 kHz parking-sensor units), in two opposed concave arrays of 36.
> - **Drive:** an Arduino Nano makes the 40 kHz square wave; an L298N H-bridge drives the two arrays in antiphase; ~10–20 V supply — *low voltage, genuinely safe*.
> - **Structure:** 3-D-printed spherical-cap holders.
> - **Cost:** ≈ $70. Designed explicitly to be built at home or in a classroom.

What it lifts: expanded-polystyrene beads up to ~3–4 mm, water and solvent droplets, small electronic parts, and small insects (ants, flies) — i.e. **milligrams up to a few hundred milligrams**, ant-to-pea scale. The focus runs at roughly 150–170 dB SPL. This is a real, reproducible, slightly magical thing to have on a desk.

2.2 Near-field acoustic levitation (the cymbal regime)

A second, heavier-duty acoustic mode underlies the hoverbike companion. In *near-field acoustic levitation* (NFAL) a flat object floats on a thin squeeze-film of trapped air — tens of micrometres — just above a surface vibrating ultrasonically. The trapped film's time-averaged pressure exceeds ambient, and the object rides it.

> **Documented** · NFAL is used industrially for contactless bearings and silicon-wafer handling. Demonstrated squeeze-film load capacities reach the order of **~0.8 N** for suitable actuators — far more than a standing-wave trap — *but only microns above the surface and only for flat, light parts*. It is a proximity effect, not free flight.

2.3 The honest scaling ceiling

Here is where a builder is owed the truth, because it is an order-of-magnitude wall, not an engineering inconvenience. Radiation pressure is bounded by the acoustic energy density, and the energy density of a sound wave in air is hard-capped by the nonlinear limit: a wave cannot rarefy below vacuum, so the maximum sinusoidal pressure amplitude is about one atmosphere — ≈ 194 dB SPL, an intensity of $\approx 2.5 \times 10^7$ W/m^2. Push past that and the wave shocks into a blast wave.

$$p_{max} \approx 1 \text{ atm} \approx 10^5 \text{ Pa} \Rightarrow \sim 194 \text{ dB SPL} \Rightarrow I \approx 2.5 \times 10^7 \text{ W/m}^2$$

> **Physics verdict · Human-scale free-air acoustic levitation is excluded, not merely hard.** Two independent walls: (1) stable trapping requires the object to be *smaller than half a wavelength* (~4 mm at 40 kHz); a person spans thousands of nodes and antinodes and would be shaken, not lifted. (2) Sustaining anything near the 194 dB ceiling over a body-sized region means megawatts of acoustic flux — lethal by internal heating and lung and eardrum rupture long before it lifts you. The radiation-pressure physics gives milligrams-to-sub-grams in free air, full stop.

2.4 What this means for the cymbal hoverbike

The companion `diy_hoverbike_paper.html` describes four bronze cymbals lifting 150 kg on ~90–300 W. The honest reading of the numbers above: that figure is off by many orders of magnitude, and the paper 's own §4 quietly concedes it — the first-light bench shows the lid "deflect…by a few microns to a few millimetres… conventional acoustic physics; no new effects are required." That is the real measurement, and it is gram-scale.

> **Open thread ·** This is not a reason to scrap the cymbal rig — it is a reason to be precise about its job. The twelve-piezo cymbal/pot-lid build is an excellent *demonstrator* of category-1 lift and, more importantly, it is the exact apparatus the framework needs to test category 2: a coherent, high-Q, asymmetrically-driven body on a pendulum, instrumented for a force that conventional acoustics does *not* predict. The acoustic lift you will measure is small and real; the x_v question is whether anything rides on top of it. Keep the bench; demote the 150 kg headline to "aspirational, pending a category-2 receipt."

§3 The high-voltage track — the ion lifter

This is the “simple high-voltage thing,” and it is the device at the center of the entire electrogravitics legend (§4). It is also, on the physics, the most over-interpreted object in the field — so building one and measuring it is the best inoculation against a century of confusion.

```
      thin corona wire  (emitter, +20 to +40 kV)
      ........................
                   ↑ ion wind ↑         gap d ≈ 1 mm per kV
      ────────────────────────
      ║        aluminium foil skirt       ║   (collector, grounded)
      ║        on a balsa triangle         ║
      ════════════════════════
   net thrust ↑ = reaction to the downward ion wind
```

Figure 2. The asymmetric capacitor (“lifter”). Corona at the sharp wire ionizes air; ions drift to the skirt, dragging neutral air with them; the frame recoils upward. Tethered — the heavy high-voltage supply stays on the ground.

3.1 The physics, stated correctly

At the thin wire the field exceeds the corona threshold of air (~30 kV/cm) and ionizes it. Ions drift down the field to the wide skirt, colliding with neutral molecules and handing them momentum — an *ion wind*. The frame recoils upward. The one-dimensional ideal:

$$T \approx I \cdot d / \mu$$

where T is thrust, I the corona current, d the electrode gap, and μ the ion mobility of air ($\approx 2\times10^{-4}\ m^2V^{-1}s^{-1}$). Two consequences a builder must internalize: thrust scales with **current×gap, not voltage** (voltage only has to clear corona onset and stay below spark-over), and thrust-per-current is small — at a 3 cm gap, ≈ 0.15 mN per milliamp. Currents are microamps to a few milliamps, so the lift is **grams-force**.

3.2 The buildable device

Build it now · Classic JLN-Labs-class lifter.

- **Frame:** balsa, equilateral triangle, sides ~10–40 cm.
- **Collector skirt:** aluminium foil band ~2–4 cm tall on the lower edge, top edge smooth/rounded (sharp edges cause back-corona).
- **Emitter:** bare ~0.08–0.1 mm wire (38–44 AWG) strung parallel above the foil.
- **Gap:** rule of thumb ~1 mm per kV — at 30 kV, ~30 mm.
- **Mass:** a 10 cm-side lifter ≈ 2 g. Tethered; only the airframe flies.

- **High-voltage supply:** a salvaged CRT/TV flyback + ZVS or 555 driver → ~10 kV AC, then a Cockcroft–Walton multiplier ladder (1 nF/30 kV caps, ~$2.50 each) → ~30 kV DC. Total ~$30–80. A series high-voltage resistor (hundreds of kΩ–MΩ) current-limits faults.

Performance: lifts its own ~2–5 g, tethered, indoors, in still air. This is the genuine article — a heavier-than-air craft with no moving parts, hovering on your bench.

3.3 The legitimate milestone — MIT's solid-state plane

Documented · Xu, He, Barrett *et al.*, "Flight of an aeroplane with solid-state propulsion," *Nature* 563, 532–535 (2018): the first sustained heavier-than-air flight with **no moving parts**. Mass ~2.5 kg; wingspan 5 m; ~40 kV; flew ~60 m in ~10 s across a gymnasium, ten-plus flights. Simultaneous thrust density **3 N/m^2** and thrust-to-power **6.25 N/kW**. This is the peer-reviewed ceiling of what EHD thrust does, and it is real.

A caution you will meet in the wild: the widely-quoted "110 N/kW, better than a jet's 2 N/kW" figure is from MIT's 2013 *static bench* thrusters at very low thrust density — true, but not a flying-vehicle number. Cite it only with that caveat.

3.4 Safety — this one can kill you

Safety — real, not decorative · Twenty to forty kilovolts is lethal, not theatrical.

- **One-hand rule.** Keep one hand in a pocket; never let current cross your chest. A path hand-to-hand through the heart can fibrillate it.
- **Discharge before touching.** Multiplier capacitors hold charge after power-off. Short the output to ground with a grounded discharge stick, every time.
- **Current-limit the supply.** A μA-limited 30 kV source is survivable in a way a stiff low-impedance one is not. Use the series HV resistor.
- **Ozone.** Corona makes it; it is toxic; ventilate.
- No jewelry, dry hands, one person, isolated area. There is no GFCI at these voltages.

3.5 The vacuum truth, and why it won't carry a person

The lifter feels like antigravity. It is not. It is a fan with no blades, pushing on air. Three walls, each fatal to the "scale it up" dream:

Physics verdict · (1) It dies in vacuum. Ion *wind* needs neutral air to push against; remove the air and the thrust goes to essentially zero. (Ion *thrusters*, which work in space, expel the ions themselves as reaction mass — a different machine.) **(2) Thrust density is tiny.** At the best demonstrated 3 N/m^2, one person (~1000 N) needs ~330 m^2 of electrode — before the batteries and high-voltage gear, which dominate the mass. **(3) Power scales brutally.** Even at 6 N/kW, 1000 N of lift draws ~170 kW continuously, far beyond any flyable battery+HV budget. Air's ~30 kV/cm breakdown caps the field, so you cannot buy thrust density with more voltage — you just arc. Human-carrying EHD lift is not on the table with known physics or materials.

PART II — SIGNAL IN THE FRINGE: THE REDACTED SCIENTISTS

§4 Townsend Brown & electrogravitics

Thomas Townsend Brown (1905–1985) is the origin of the whole "antigravity went black in the 1950s" story. He deserves a careful read, because his record contains real documents, a real effect, and a real error — and the legend conflates all three.

Documented · The documented man. Caltech and Denison attendance; U.S. Navy and Naval Research Laboratory service from 1930; participation in the 1932 Navy–Princeton gravity expedition. Real granted patents, including GB 300,311 (1928), US 2,949,550 "Electrokinetic Apparatus" (1960), US 3,018,394 "Electrokinetic Transducer" (1962), and — note for Part III — US 3,022,430 "Electrokinetic *Generator*" (1962). Project Winterhaven (1952), his proposal for a Mach-3 electrogravitic craft, is archived at the University of Maryland. The 1929 *Science and Invention* article "How I Control Gravitation" is a real publication. His dated laboratory notebooks (1955–77) and the 1952 ONR audit are read in full and scored in the companion paper Townsend Brown, which also traces the "time travel" lore attached to his name to its two sources (Moore–Berlitz 1979; Schatzkin's anonymous "Morgan") — neither of them Brown.

Proponent's claim · Brown asserted the force on his charged "gravitator" was an *electrogravitic* coupling — a genuine interaction between charge and gravity, the same effect that would later steer a saucer. The "Biefeld–Brown" name credits a Denison professor as co-discoverer; present-day Denison University says it has *no record* of the experiments or the association. Treat the mentor story as Brown's account.

> **Physics verdict · The asymmetric-capacitor force in air is ion wind, and the published vacuum tests bound the *static* effect.** Talley (1988, Air Force-funded) ran Brown-style electrodes at 10^{-6} torr under DC and measured *no thrust* except during actual breakdown current. Tajmar (2004) isolated the corona wind directly. The load-bearing modern result is Tajmar & Schreiber, *Acta Astronautica* 181, 482 (2021): a high-voltage capacitor on a balance to 10 kV, sensitivity ±0.3 mg, **no force detected** in any configuration — ruling out the proposed electrostatic-propulsion theories by many orders of magnitude; the 2024 Dresden sweep (Tajmar, Kößling & Neunzig, *Sci. Rep.* 14, 19427) extends the null across capacitor geometries, solenoids and tunnelling currents at nano-newton resolution. The atmospheric effect is electrohydrodynamic, not gravitational. It is §3, dressed by others in mythology — and Brown himself got there first: his 1973 notebook re-appraisal concludes the torque is "a direct function of the total current. There is no other energy source," and his 1958 theory predicts a *minimum* force in vacuum. The one calibrated measurement of his saucer, made on his own apparatus by the Navy's auditor (Cady, ONR 1952: ~7 g at 30 kV, ~1 g with the corona wire shrouded), found electric-wind theory "satisfactory in order of magnitude" and the apparatus "well engineered and extremely reliable." He was a careful experimenter who mis-attributed a real force. He did claim vacuum operation — in letters (1955, 1973), from memory, without notes — but not in his notebooks, and the French torsion balance he cited was, now that the report has been read, null on his own geometry.

Two scoping notes keep that verdict exact rather than merely emphatic. **First, the one claimed vacuum positive.** Brown's 1955–57 work with SNCASO in Paris ("Project Montgolfier") is a documented corporate program whose archive Cornillon released before his death in 2008, and whose two reports (1 Jul 1957; final 15 Apr 1959) were read in the original on 2026-08-18 (`MONTGOLFIER_REPORT_RETRIEVAL_2026-08-18.md`). The 1957 report attributes the *air* motion to a non-polarised electrode effect — electric wind, agreeing with the paragraph above — and its bell-jar vacuum rotations were polarity-locked ("always toward the positive pole"), often started only by internal flashes, killed by a grounded screen, and unmeasurable; Cornillon himself proposed an "anodic effect" (particles torn from the anode) as the cause — the vacuum-arc-momentum reading (Martins & Pinheiro 2011). The 1959 final report, in a 1.48 m chamber at 5×10^{-5} mm Hg on a torsion balance at ±50–55 kV, found the Brown disc–wire article *null*, two other encapsulated articles *null*, and one $BaTiO_3$-loaded capacitor giving 30° ≈ 1 g·cm, reversing with polarity — with no current, breakdown, or lever-arm record. So the honest sentence is: every *published, independently instrumented* vacuum test is a null; the French program's own balance was null on Brown's geometry; and its one positive is a single uncontrolled sponsor's number on a high-K article, open until a bench with a simultaneous current-and-force log speaks. **Second, what the nulls cover.** Every test article in the null program is a static, incoherent capacitor. That is exactly the object this paper's companion framework predicts to couple at the bare Einstein-constant floor ($\sim 10^{-43}$) — the nulls are the framework's own retrodiction (scored in `RESULTS_BIEFELD_BROWN_2026-08-13.md`), not its refutation. And the underlying mechanism is not in doubt at all: electromagnetic fields curve spacetime by theorem (Einstein–Maxwell, solved exactly for real coils by Füzfa, *Phys. Rev. D* 93, 024014, 2016). What Brown had was the right shape with the wrong active ingredient — high voltage on an incoherent dielectric — and a magnitude problem forty orders deep. One scoping sentence the verdict needs: every positive in the vacuum record (Brown's 1956 impulse, the Montgolfier flashes and its $BaTiO_3$ article, Talley's breakdown/pulsed forces, Buhler's stacks) is *transient*, which is both the regime a vacuum-arc discharge explains and the only regime the framework's own force law allows — so the lineage is refuted as a static effect and open as a transient one, pending the four discriminators in Electrogravitics §1.1. The engineering reading of that lineage is now the job of that paper.

One honesty flag, because proponents lean on it: the U.S. Army Research Laboratory analysis (Bahder & Fazi, 2003) is often cited as if it supported electrogravitics. It did the opposite in spirit — it found a naïve "breeze" estimate too small but full ion-drift momentum the right magnitude, and explicitly called for the vacuum test that Tajmar later answered with a null. It is an ion-wind paper.

4.1 "It went black" — document vs. inference

> **Documented ·** The report is real: *Electrogravitics Systems* (Aviation Studies (International) Ltd., February 1956) names Glenn L. Martin, Convair, GE, Bell, Sperry-Rand and others as interested in electrogravitic propulsion and proposes a disc interceptor.

> **Open thread ·** But "named in a 1956 think-tank survey" is not "ran a funded program that produced hardware and then classified it." The disappearance of electrogravitics from the open literature after ~1957 is explained at least as well by the ion-wind debunking — there was no gravity effect to report — as by secrecy. The suppression reading is *possible*; it is not *documented*. This is the cleanest place in the whole field to feel the difference between the two.

4.2 The genuine open thread — Buhler's vacuum claim

> **Proponent's claim ·** Charles Buhler (a NASA veteran) and Exodus Propulsion Technologies report an asymmetric *dielectric* capacitor producing net thrust *in vacuum* — figures around

10 mN in a stack, across >2000 runs, presented at APEC in December 2023 (patent US 11,511,891). If real, it would be the thing Brown was wrong to claim but right to chase.

Physics verdict · Status: **not peer-reviewed, not independently replicated.** The results live in conference talks and patents. Sub-Newton forces are exactly where thermal, outgassing, and electromagnetic artifacts hide — the same trap that ultimately explained the EmDrive (which Tajmar also tested to null). This is a real open thread, not a receipt. The decisive moves — independent replication, journal review, a space demonstration — are outstanding. *Watch it; don't bank on it*. Since this was written the claimed force has fallen roughly forty-fold — from "1 g on a 30–40 g device" (April 2024) to "5–10 mN" under a March 2026 rebranding as the "electrostatic pressure force" — with no published explanation, and still no paper or independent instrumented test. A claimed magnitude that shrinks as scrutiny grows is the signature Tajmar documented for the EmDrive and the Mach-effect thruster; note it, and keep watching.

§5 Ning Li & the superconductor-gravity program

If electrogravitics is the high-voltage lineage, the rotating superconductor is the cryogenic one — and it is the better-pedigreed of the two, with real *Physical Review* theory and real NASA replication attempts. It also contains the most important honest correction in this paper, because the Ning Li story is told almost everywhere as a disappearance, and it was not one.

Documented · **Ning Li (宁李, 1943–2021)** was a research scientist at the University of Alabama in Huntsville. With Douglas Torr she published a real theory predicting that a rotating type-II superconductor — its Cooper-pair condensate in coherent rotation — would generate an anomalously large gravitomagnetic field: Li & Torr, *Phys. Rev. D* 43, 457 (1991); *Phys. Rev. B* 46, 5489 (1992); Torr & Li, *Found. Phys. Lett.* 6, 371 (1993). In 1999 she left to found **AC Gravity LLC**, which received a U.S. Department of Defense award of **$448,970 (2001)**.

Physics verdict · The theory was refuted on its own terms: Harris, *Found. Phys. Lett.* 12, 201 (1999), showed the predicted fields were too large by many orders of magnitude under realistic assumptions. AC Gravity LLC published no results. The often-quoted "11–12% weight loss" is *not* Li's — those weight-change numbers belong to Podkletnov (below), and Li's own 1997 experiments failed to reproduce the effect. Keep them separate.

Documented · **The correction the conspiracy frame gets wrong.** Ning Li did not "vanish" or get "disappeared back to China." Per her obituary, a 2014 vehicular accident caused permanent brain damage, followed by Alzheimer's; she died in Huntsville on **27 July 2021, age 78**. The mystery was a tragedy, not a cover-up — and saying so is exactly the discipline that makes the genuinely open threads credible.

5.1 Podkletnov and the rotating disc

Proponent's claim · Eugene Podkletnov (Tampere University of Technology) reported ~0.05% weight loss of a mass suspended above a rotating YBCO superconducting disc, rising toward ~0.3–2% under AC fields (Podkletnov & Nieminen, *Physica C* 203, 441, 1992). A 1996 follow-up was withdrawn amid press leaks and his institution distanced itself. Later he claimed an "impulse gravity generator" emitting a collimated repulsive beam (Podkletnov & Modanese, arXiv physics/0108005, 2001).

Physics verdict · NASA Marshall took it seriously enough to try — the Koczor/Noever program, and a careful replication by Hathaway, Cleveland & Bao (*Physica C* 385, 488, 2003) — and found **no weight change to the 0.001% level**. A German Göde-Foundation replication of the impulse generator returned null. After thirty years the effect remains unreplicated. That is not proof it is impossible; it is the absence of the receipt that would make it real.

5.2 Tajmar — the man who tested his own claim to death

Documented · Martin Tajmar is the figure to anchor on, because he ran the arc honestly. In 2006 he and de Matos reported a "gravitomagnetic London moment" — an accelerometer signal around a niobium ring spun to thousands of rpm (arXiv gr-qc/0610015), publicized by ESA. By 2008, with no independent reproduction, **Tajmar himself** traced the signal to thermal/cryogenic and electromagnetic artifacts. He went on to run the rigorous vacuum null on the Brown effect (§4) and on the EmDrive.

Thesis sentence · *Tajmar is what the field looks like when it works: a serious person builds the apparatus, reports the positive, fails to replicate it, and says so in print. Every name in this section was chasing a real question. The ones who advanced it are the ones who let the null count.*

§6 Amy Eskridge

You asked specifically about Amy Eskridge, and she deserves to be handled with care — she was a real person who died young, and her name now travels mostly as a data point in a viral "dead scientists" list. The honest account is sadder and smaller than the legend, and worth stating plainly.

Documented · What is documented. Eskridge was an aerospace/propulsion figure in Huntsville with a UAH science background. She co-founded and chaired the **Institute for Exotic Science**, a small Huntsville organization oriented to advanced propulsion, zero-point energy, and "antigravity" — co-founded with her father, **Richard Eskridge, a retired NASA Marshall plasma-physics engineer** (the real NASA link runs through him, not through her own employment). She gave a dated public talk, "A Historical Perspective on Anti-Gravity Technology," to the Huntsville L5 Society in December 2018 — a survey, not a device demonstration. She died on **11 June 2022, age 34**; the ruling was a self-inflicted gunshot wound.

Physics verdict · What is not documented: any working device, any peer-reviewed result, any confirmed Woodward/Mach-effect project, or that she was personally a NASA employee. The Institute's output is a topic space and a talk, not a receipt. There is a real evidentiary floor here, and it is low.

Proponent's claim · The suppression narrative — the reported "if you see a report that I killed myself, I most definitely did not" message, and her inclusion in 2025–26 viral "dead scientists with access to U.S. secrets" coverage — is real as *reporting*, but unsubstantiated as *fact*. Her own father stated he did not consider the death suspicious and attributed it to chronic pain. Authorities have indicated no connection between her work and her death.

Open thread · The honest signal in the Eskridge story is not a hidden antigravity device. It is that the antigravity community is small, under-funded, emotionally intense, and prone to reading tragedy as conspiracy — and that a real young engineer's death got absorbed into that pattern. The respectful and the rigorous move are the same one: hold the documented facts, decline the unsupported frame, and don't build a physics claim on a person's death.

§7 Tesla & energy from the earth

The recurring rumor — that these craft are also free-energy machines, "drawing power from the earth" the way Tesla promised — is where the most beautiful real physics sits next to the most durable myth. The job here is to keep the two from trading clothes.

> **Documented · What Tesla actually built and patented.** At Colorado Springs (1899) he built a "magnifying transmitter" and recorded what he called terrestrial stationary waves — concluding the Earth behaves as a resonant conductor. Wardenclyffe Tower (1901–1917) was meant to transmit power and telegraphy wirelessly; it was never completed and was dismantled in 1917. The real patents: US 645,576 and 649,621 (1900, transmission via Earth and upper atmosphere); US 1,119,732 (1914, the magnifying transmitter); and US 685,957 / 685,958 (1901, "Apparatus for / Method of Utilizing Radiant Energy").

> **Physics verdict ·** Read the radiant-energy patents for what they say: an elevated insulated plate charges a condenser from incident radiation, which a discharge circuit then taps. It is a **charge-accumulation device — it gathers and stores incident energy; it does not create it.** Wireless power transfer is real (and radiatively lossy over distance); "free unlimited energy from the aether" is neither demonstrated nor what the patents claim. The famous "200 lamps lit 26 miles away" is Tesla's own narration, not an independently verified demonstration.

> **Documented · The real referents under the myth.** The Earth–ionosphere cavity has a genuine fundamental resonance — the **Schumann resonance, ~7.83 Hz** (harmonics ~14.3, 20.8 Hz), driven by global lightning. Predicted by Schumann (1952), first measured by Balser & Wagner, *Nature* 188, 638 (1960). Tesla anticipated Earth-cavity resonance qualitatively ~50 years early — a real insight he then over-read. The global atmospheric electrical circuit is also real: a fair-weather field of **~100 V/m** at the surface, an ionosphere–Earth potential of **~250 kV**, total circuit current **~1–2 kA**.

> **Physics verdict ·** But you cannot cheaply tap it. The fair-weather conduction current density is ~2 picoamps per square metre and the source impedance is gigaohms-plus: the energy density of the air is minuscule. Real atmospheric-electricity harvesting exists in the lab and produces negligible power. The earth has energy; it does not have *free, concentrated, low-impedance* energy waiting at the top of a tower.

§8 Schauberger & implosion

Documented · Viktor Schauberger (1885–1958) was an Austrian forester and close observer of moving water — log-flume transport systems, river vortices, trout holding station in fast current. Those observations are real and were genuinely ahead of their time. The full record — life, machines, witnesses, physics — is the companion paper Viktor Schauberger; this section is its ledger entry.

Proponent's claim · From them he built "implosion" theory: that inward, centripetal, vortical flow could purify water, generate energy, and — in the "Repulsine" disc — produce lift. Wartime accounts tie a prototype to SS-directed work; he went to the United States in 1958 to commercialize implosion, signed away his work, and died days after returning to Linz.

Documented · The machine is real, and its physics is ordinary. Priced against fluid mechanics (companion §9.0), the Repulsine is a bladeless *corrugated centrifugal rotor* — two copper wave-plates forming a narrow whorl gap, central intake, rim exhaust over a domed shell; its nearest peer-reviewed cousin is the Tesla disc turbine. Its lift channel is the same Coandă/centrifugal-jet mechanism as the Avrocar and every Coandă UAV: a 0.6 m rotor at 10–20 kW gives 15–60 kgf. Its intake genuinely chills and fogs ($v^2/2c_p$; Ranque–Hilsch separation); its glow is genuine spray/dust electrification and corona, and the *green* he reported is the copper spectrum. So a light copper unit jumping off its mount, a cold wet intake, a hot rim, and a green flicker are all what ordinary turbomachinery does. The water-side residue is now peer-reviewed: Walter Schauberger's hyperbolic-funnel vortex out-aerates mechanical aerators (*Water* 2022, 2025; *Phys. Fluids* 2024), and the *gravitational water-vortex power plant* (Zotlöterer) is a lawful low-head turbine at 15–30 % (up to ~60 % optimised).

Physics verdict · What was never measured, and what is unsupported. No thrust, power, temperature or rpm figure from any original device exists, friendly or hostile; the only instrumented replicas (Implosion e.V., on the original PKS dies) show torque curves, membranes deforming above 7,000 rpm, and no levitation. The flight accounts trace to Schauberger's own 1941 letters and a 1956 letter (companion §5.1) — contemporaneous, sincere, and with *zero named first-hand witnesses*; the family institute itself says the saucer claims "cannot be verified." The magnitude claims are unsupported by orders: "228 t from a 20 cm disc" exceeds the aerodynamic bound ~2,700×; a 0.6 m copper rotor cannot survive the 10–20 krpm claimed (hoop stress 2–4 GPa vs 0.07–0.3 GPa yield); "self-running after the motor is cut" and "net cooling of the room" fail the second law as closed systems and have never been observed. *This paper's earlier stamp, "pseudoscience," is withdrawn*: it named neither the real machine nor the real gap. The register is **real machine, ordinary lift physics, never instrumented; anomalous residue unsupported** — and the bench question is narrow: the wave-disc rotor at 20 cm on a metered motor, a load cell, and an intake thermocouple.

Proponent's claim · **The framework's interest** is architectural, not performance: a spun, charged, cold-vortex copper shell assembles four of the warp drive's named levers in one device (companion §9, "Architecture C"), and the open-system reading of "implosion energy" (companion §9.6) is lawful *net-extraction unproven*, not over-unity. Neither is a receipt.

§9 Over-unity, in general

The unifying claim across Brown's self-powered disc, Searl's generator, Otis Carr's self-recharging "Utron" saucer, Schauberger's *self-running* implosion machine, and the "it makes its own energy" rumor is the same one — and this network deliberately does *not* file it under "forbidden by thermodynamics." Reasoned from first principles, energy accounting over *device + reservoir* is an identity; the only physical content of "conservation" is *which reservoir*. So every such claim is put through the same three questions the energy lane uses (`zero_point_energy.html` §7, `the_vta.html`): **is a reservoir named? what does the coupling price to? does the device run cold under load?** A device that names its reservoir and shows the cold signature has earned a bench date; one that names neither is a *bookkeeping* contradiction — a closed loop claiming more out than in — which is the only sense in which anything below is "forbidden." (Schauberger's *lift* claim is a different object — §8 — and does not belong in this paragraph.)

Physics verdict · *Closed-system* over-unity — cyclic net work from a single equilibrium bath with no second reservoir named — is a bookkeeping contradiction, and that is the whole of what the first and second laws say. *Open-system* extraction from a named reservoir is ordinary (a heat pump; a solar cell; the measured Moddel-class Casimir-cavity cell, ~70 W/m^2, output $\propto$ 1/d, peer-reviewed) and this network treats the vacuum sector as such a reservoir. The historical pattern in the saucer literature is nonetheless invariant: a device that shines in open-loop or single-run demonstration fails the closed-loop, independently-instrumented measurement, and none has ever reported the one signature that would separate extraction from artifact — running *cold* under load. The **Searl Effect Generator** — a magnetized rotating-ring device claimed to produce free energy *and* antigravity — has never been observed or replicated under controlled conditions and is not a recognized principle. An "antigravity device that also powers itself" is not two extraordinary claims; it is one impossible one, because a closed system cannot output more work than it takes in.

Documented · **The 1950s type specimen: Otis Carr's OTC-X1 (1957–61).** The best-documented case of the package deal, and worth one paragraph because it keeps being revived (a 2021 fringe-journal paper derives 10^{13} W and 10^{5} T from it). The record: the only patent (US 2,912,244, 1959) is, by its own nine claims, an amusement ride; the 6-ft prototype failed on television film under an *external* motor, leaking its mercury; the SEC enjoined the stock and a jury assessed the statutory-maximum $5,000 fine (no prison term — the "14 years" online is a garble). The physics: Carr's "580 rpm on 45 ft" decodes as rim speed ≈ Earth's equatorial 465 m/s, where an aluminium hull is 1.7× past yield and Earth's own centrifugal relief is 0.35% of g; the "Utron" is a mercury homopolar disc delivering ∼8 V; the revival's 10^{5} T rests on counting every conduction electron as current (10^{14}× over its own Hall estimate; 1.9 Mt of field energy per m^3); and matched counter-rotation is exactly the configuration in which any coupling cancels. Two honest residues survive — liquid-metal homopolar machines and self-excited liquid-metal dynamos are real, as *loads* — and are filed in the companion Otis T. Carr and the OTC-X1, which also explains why Carr is the wrong poster child for any suppression story: the file is thick and points the other way.

Documented · **Where the real vacuum energy lives, and what it actually pays.** The Casimir force between close conductors is real and measured — Lamoreaux (1997), Mohideen & Roy (1998) — and the dynamical Casimir effect (photons from the vacuum via an accelerated mirror) was observed by Wilson *et al.*, *Nature* 479, 376 (2011); in both, the energy is paid by external work. The measured *net-output* anchor is different and newer: Moddel-class optical Casimir-cavity MIM cells (Moddel et al., *Phys. Rev. Research* 2021; patent US20250096702A1) deliver ∼70 W/m^2 with output ∝ 1/d, and this network's own budget paper (`casimir_energy_extraction.html`) prices what the reservoir can supply: the *stored* Casimir energy is exhausted in ∼172 ns and is not the fuel; the ambient near-field thermal sector is, at ∼10^7 W/m^2, so supply is not the constraint — *rectifiability* is, and the honest lane-status is "open-system vacuum harvesting is real at the W/m^2 scale; scaling is an engineering question about the junction." What the saucer literature does with this is the move named below: borrow the word and skip the reservoir inventory. Floyd Sweet's VTA, the most-cited example, is filed in `the_vta.html` as a mechanism scaffold with one measured anchor class behind it — explicitly *not* a claim that the VTA worked (no independent instrumented measurement survives; the honest prior on its bench is null until the frozen protocol runs).

Documented · **The honest "energy from the environment" menu** — the real set the myths distort: solar (~1000 W/m^2, the one genuinely large ambient source), RF/EM harvesting (microwatts — ~tens of $\mu W/m^2$ ambient, sensor-scale), thermoelectric and piezo harvesting (μW–mW), telluric earth-battery currents (real, tiny, powered early telegraph lines). All conserved. All modest. None over-unity.

Thesis sentence · *The free-energy mythos has one move and makes it every time: take a real effect that is small or conserved — Schumann's cavity, the fair-weather field, the Casimir force, a water vortex — and inflate it, without naming a reservoir or reading a thermometer, into a self-running machine that has never survived independent measurement. The signal is real. The unaccounted reservoir is the noise added on top. The discipline is not "conservation forbids"; it is "show me the reservoir and the thermal sign."*

PART IV — THE LEDGER

§10 Buildable / claimed / forbidden

The whole survey, on one page, in the three registers of §1.

Effect / device	Status	What the record actually shows
Ultrasonic standing-wave levitator (TinyLev)	**Buildable**	Lifts mg–sub-g (beads, droplets, insects). ~$70. Conventional acoustics.
Near-field acoustic levitation (squeeze-film)	**Buildable**	~0.8 N for flat parts, microns above a vibrating surface. Industrial.
EHD ion lifter / ionocraft	**Buildable**	Grams-force at 20–40 kV. MIT flew 2.5 kg, no moving parts (Nature 2018). Ion wind.
Human-scale free-air acoustic hover	**Forbidden**	Excluded by the ~194 dB / 1-atm energy ceiling and sub-$\lambda/2$ trapping. Lethal long before lift.
Human-scale EHD / lifter craft	**Forbidden**	~330 m^2 electrode and ~170 kW per person; zero thrust in vacuum.

Effect / device	Status	What the record actually shows
Biefeld–Brown “electrogravitics”	Refuted	Vacuum nulls (Talley 1988; Tajmar & Schreiber 2021, ±0.3 mg). It is ion wind.
Li–Torr rotating-superconductor gravitomagnetism	Refuted (theory)	Harris 1999: predicted fields too large by orders of magnitude. AC Gravity: no results.
Podkletnov gravity shielding / impulse beam	Unreplicated	NASA Marshall & Göde nulls to 0.001%. 30 years, no receipt.
Buhler “Exodus” vacuum thrust	Open thread	Claimed ~mN in vacuum; not peer-reviewed, not replicated. Watch it.
Tesla wireless power / earth resonance	Real, modest	Schumann 7.83 Hz and the ~1 kA global circuit are real; not a tappable free-energy source.
Schauberger implosion / Repulsine	Real machine, never instrumented	Bladeless corrugated centrifugal rotor; ordinary Coandă/jet lift (tens of kgf at 10–20 kW), real intake cooling, real spray electrification. Flight lore = his own 1941/1956 letters, zero named eyewitnesses. Tonnage/self-running claims unsupported by $\sim 10^3$. Residue now peer-reviewed: vortex aeration, vortex micro-hydro.
Carr OTC-X1 / Utron (1957–61; Gobbi 2021 revival)	Refuted / priced	Amusement-ride patent; prototype failed on film under external drive (mercury leak); “580 rpm” = rim speed ≈ Earth’s 465 m/s (0.35% g relief, Al hull 1.7× past yield); Utron ≈ 8 V homopolar disc; 10^{13} W revival = $10^{14}\times$ current error; spinning-mass weight nulls (Faller / Nitschke / Quinn 1990) bound the whole class. Full audit: otis_carr.html.
Self-powered disc, closed loop, no reservoir named (Searl, etc.)	Bookkeeping contradiction	More out than in with no second reservoir is an accounting error, not new physics; never replicated closed-loop; none has ever reported running cold under load. Reservoir-naming open-system claims are a different row (below).

Effect / device	Status	What the record actually shows
Open-system vacuum-sector extraction (Moddel-class Casimir MIM cells)	Real, measured, small	~70 W/m², output ∝ 1/d, peer-reviewed; the network's energy lane's anchor. Reservoir = ambient near-field thermal sector rectified against the ~0 K zero-point sector; scaling is a junction problem. Not a saucer power plant at today's numbers.
x_v lattice-coupling effect (framework)	Under test	Hypothesis with a pre-registered bench protocol. No receipt yet; allowed to stay open.

§11 The honest build program

Gathering everything leads to one program, and it is the same discipline the rest of this network already runs on its mining benches: build the real effect, instrument it past the point where the real effect can explain the reading, and let a pre-registered control battery — not enthusiasm — decide whether anything else is there.

1. **Build both demonstrators this weekend.** The TinyLev (§2.1) and the lifter (§3.2). Feel category-1 lift directly; it cures the urge to over-read it later.
2. **Measure them honestly.** Log thrust vs. current and gap on the lifter; log trapped mass vs. drive on the levitator. Confirm they sit on the conventional curves of §2 and §3. (They will.)
3. **Run the vacuum and orientation controls.** The lifter dies in a bell jar; the levitator dies without a node. Reproducing the published nulls yourself is how you earn the right to claim a positive later.
4. **Only then, test category 2.** Move to the twelve-piezo coherent-body bench of `oph_chi_nu_potlid_first_build.html` with its full controls battery (drive-off, orientation flip, phase flip, null pattern, dummy mass, amplitude-slope). A x_v receipt is a force that survives every one of those after the acoustic and EHD contributions are subtracted.
5. **Refuse the package deal.** If any result also seems to make energy, that is the signal to look harder for the artifact, not the moment to announce a breakthrough — because lift-plus-free-energy is the one combination the physics rules out.

Thesis sentence · *Two effects are real and you can hold them in your hand by Sunday. Every legend in this paper is one of those two effects, a refuted theory, an unreplicated claim, or an impossibility — and the only thing that has ever moved a claim across those lines is a measurement that survived its* *controls. Build the demonstrators. Run the nulls. Keep the x_v door open and unforced. Let the receipt speak.*

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Draft document, build guide and survey in one. The two demonstrators (TinyLev, ion lifter) work on conventional physics and lift small masses; their human-scale and vacuum versions are excluded as marked. The fringe-scientist and free-energy sections report the record and the controlled-experiment verdicts, not the author’s wishes. The x_V lattice-coupling hypothesis is explicitly held open and unproven, with its test specified in the companion pot-lid paper. **High voltage and high-intensity ultrasound are both genuinely dangerous — read §3.4 and §2.3 before building anything.**